MCU

文件: MCU.kicad_sch

POWER_SHARED

文件: power.kicad_sch

FOC_M1

文件: FOC_M1.kicad_sch

FOC_M2

文件: FOC_M2.kicad_sch

Dual independent motor framework
Common input supply + GD32F505VxTx controller
Review protection thresholds and ratings before energizing.

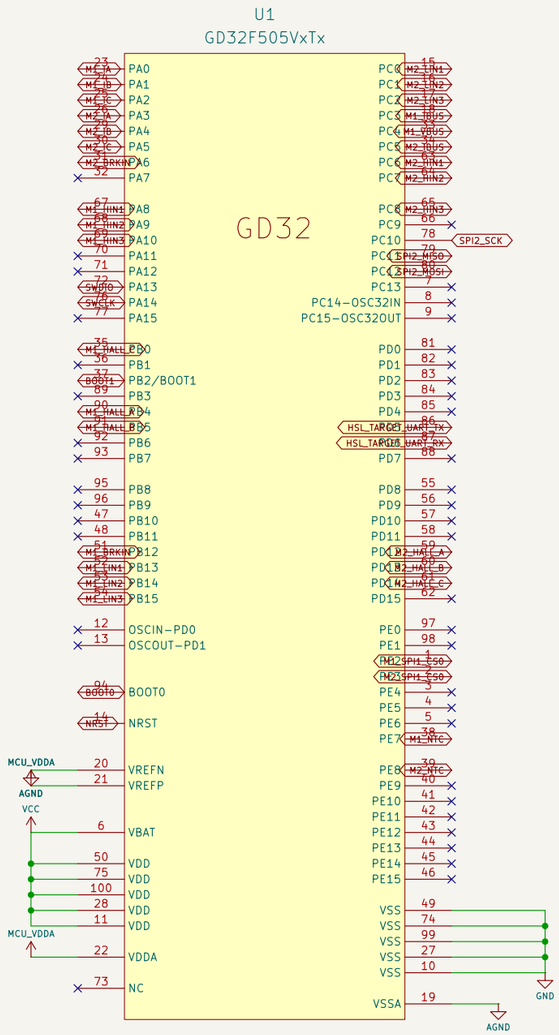
HSL_DEBUGGER

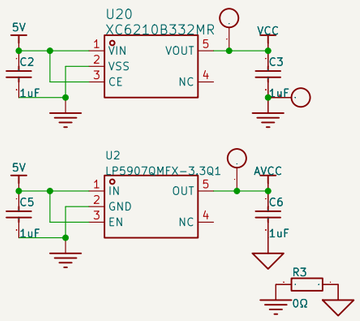
文件: HSLink.kicad_sch

Sheet: /
File: GD32F5_FOC.kicad_sch

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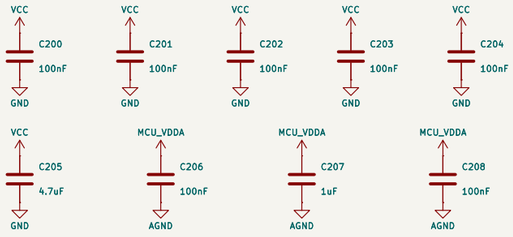
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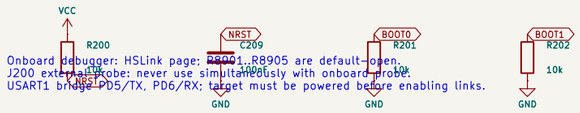














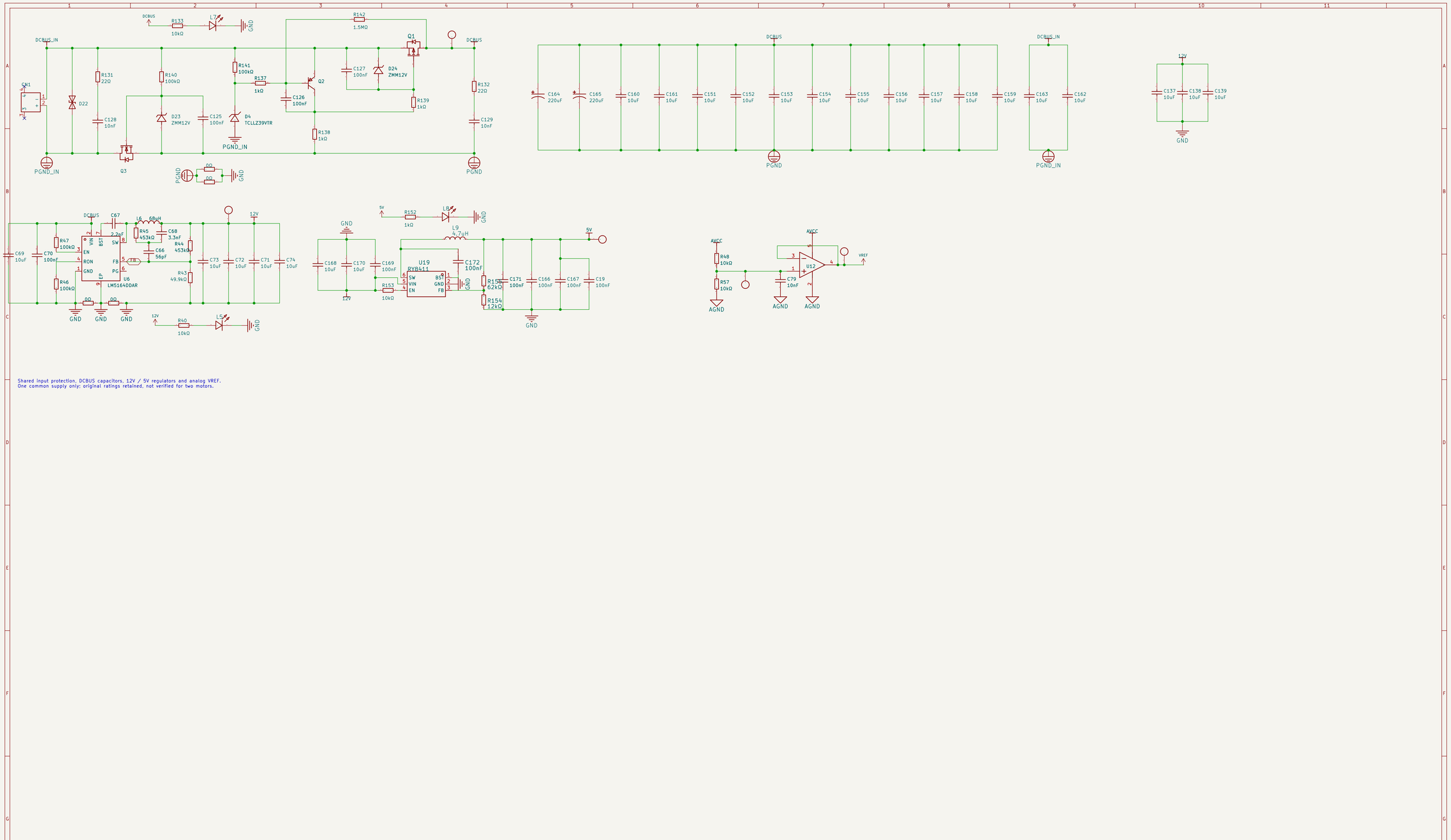


Onboard debugger: HSLink page; R8901..R8905 are default-open.
J200 external probe: never use simultaneously with onboard probe.
USART1 bridge: PD5/TX, PD6/RX; target must be powered before enabling links.

MCU_VDDA = VCC through R203 (OR), bypass C206..C208 to AGND.
AVCC LDO remains for external analog amplifiers only.
J201: 1=3V3, 2=GND, 3=M1_BRKIN, 4=M2_BRKIN.
FAULT inputs use pullups; external open-drain HEALTH holds LOW only while healthy.
Configure each timer break polarity HIGH; unplugged defaults to brake. No autonomous trip comparator fitted.
Current limits/thresholds and external fault source remain to be designed.

M1 TIMER0: PA8/9/10 AF1; PB13/14 AF2, PB15 AF1. BRKIN PB12 AF4.
M2 TIMER7: PC6/7 AF3, PC8 AF2; PC0/1/2 AF0. BRKIN PA6 AF4.
M1 currents PA0/1/2; M2 currents PA3/4/5 (ADC0/ADC1).
IBUS: PC3 / PC5; VBUS PC4; NTC PE7 / PE8 (ADC2).
Hall M1 PB4/5/0 TIMER2; M2 PD12/13/14 TIMER3. Release PB4 JTAG, keep SWD.
Shared SPI2 PC10/11/12 AF2; GPIO chip selects PE2 / PE3.
VA/VB/VC/COM and M2_VBUS stay on power-sheet testpoints, not MCU ADC.

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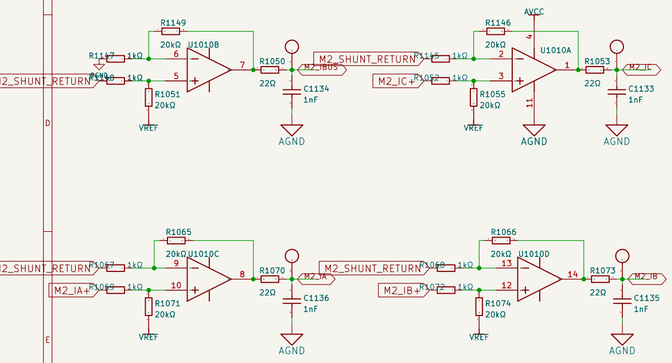
Shared input protection, DCBUS capacitors, 12V / 5V regulators and analog VREF.
One common supply only; original ratings retained, not verified for two motors.

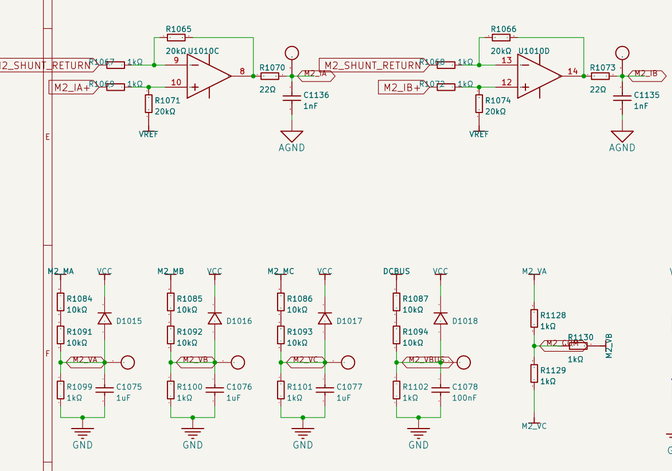
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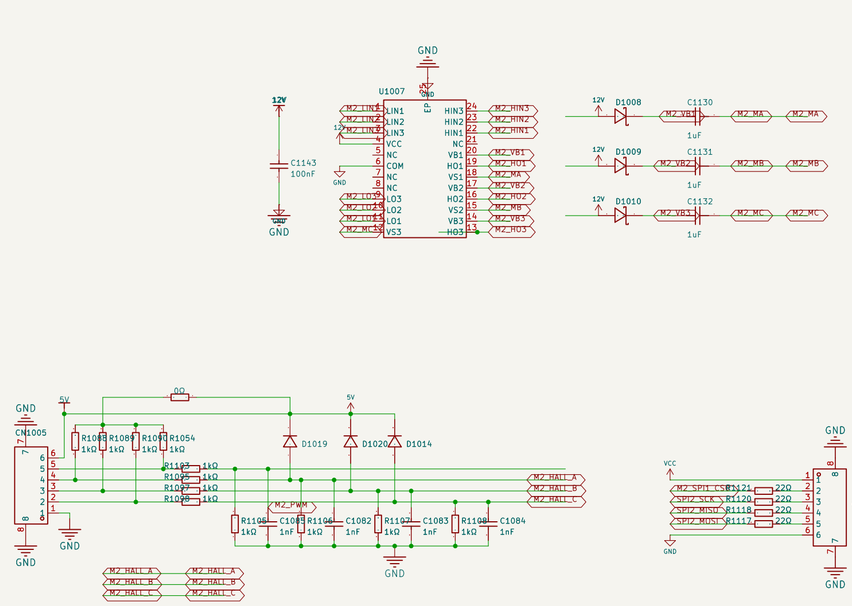
MOTOR 1: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M1_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

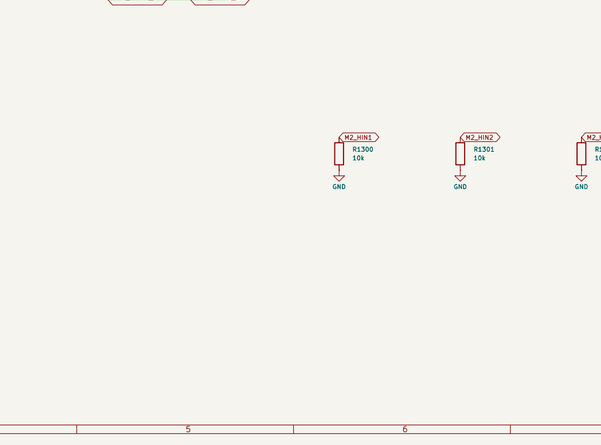
VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

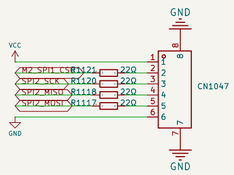
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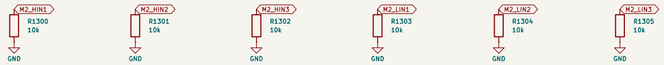












MOTOR 2: Independent gate drive, six MOSFETs, phase shunts and bus shunt.
All motor signals M2_*. Common DCBUS / 12V / 5V / VCC / AVCC / VREF.
PGND is common AFTER the dedicated bus shunt; phase-shunt return remains local.
Reference values and footprints require engineering review; no rating verification.

VA/VB/VC, COM, IBUS, NTC and auxiliary PWM are retained reference interfaces.
Only nets explicitly connected on MCU sheet are MCU functions.
FD6288Q has no dedicated EN pin; hardware gate-input pulldowns are provided.
BRKIN requires external fault generation; software must initialize PWM disabled.

Sheet: /FOC_M2/		
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HPM5301 REFERENCE DEBUGGER / HSLink-style hardware
USB-powered debugger; target remains powered by board supply.
Reference GPIO mapping retained. Firmware compatibility NOT verified.

TARGET LINKS R8901..R8905: DEFAULT DNP / OPEN
Fit 0R only with target VCC present. Remove before external J200 probe
or debugging with target unpowered. NO automatic backfeed isolation.
SWDIO / SWCLK / NRST / target TX / target RX (top to bottom).
J200 and onboard debugger must never drive target simultaneously.

H8001 pin1 = target VCC reference ONLY: no target power injection.
No validated ADC voltage sensing or firmware power-off tri-state.
H8001 other signals remain debugger side of target disconnect links.
HSL_TARGET_UART_TX = GD32 PD5 USART1_TX AF0 -> probe RX.
HSL_TARGET_UART_RX = GD32 PD6 USART1_RX AF0 <- probe TX.

Reference: HPM 4_DAP + isolated 3_POWER 5V/3.3V block.
USB1 / EtherCAT / HPM main-controller circuit intentionally excluded.
USB footprint and firmware require validation before PCB fabrication.

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